

Threat Hunting

1. Introduction

This task focuses on threat hunting using Windows Security logs, specifically Event ID 4624 (successful logon), to identify suspicious authentication activity and potential unauthorized access.

2. Objective

To detect abnormal or unauthorized successful logins by analyzing Event ID 4624 and identifying suspicious patterns such as unusual logon types, unknown source IPs, and abnormal login times.

3. Tools

Event Viewer

Windows Security Logs

4. Hypothesis

An attacker who has obtained valid credentials may successfully log into the system, generating Event ID 4624 logs. Suspicious logins may be identified based on unusual logon types, unknown source IP addresses, or abnormal login times.

5. Methodology

Step 1: Open Logs

Open Event Viewer → Windows Logs → Security

Step 2: Filter Logs

Filter Current Log → Enter Event ID: 4624

Step 3: Collect Relevant Fields

- Account Name
- Logon Type
- Source IP Address
- Timestamp

Step 4: Identify Suspicious Indicators

Logon Type 10 (Remote Desktop)

Logins at unusual hours

external IP addresses

High-privilege accounts



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6. Evidence

Timestamp	User	Logon Type	Source IP	Notes
2026-04-06 01:30 PM	Administrator	10	192.168.1.102	Suspicious RDP login
2026-04-06 01:32 PM	User1	2	Local	Normal activity
2026-04-06 01:35 PM	User1	5	10.0.2.15	Logon Successful

The screenshot displays the Windows Event Viewer interface. The left-hand pane shows the navigation tree with 'Event Viewer (Local)' selected, and 'Security' expanded under 'Windows Logs'. The main pane is divided into two sections. The top section, titled 'Security - Number of events: 12,206', contains a list of events. The bottom section, titled 'Event 4624, Microsoft Windows security auditing', shows the details for a specific event.

Event List:

Keywords	Date and Time	Source	Event ID	Task Category
Audit Success	4/6/2026 1:50:28 AM	Microsoft Windows security auditing	4672	Special Logon
Audit Success	4/6/2026 1:50:15 AM	Microsoft Windows security auditing	4672	Special Logon
Audit Success	4/6/2026 1:50:15 AM	Microsoft Windows security auditing	4624	Logon
Audit Success	4/6/2026 1:50:15 AM	Microsoft Windows security auditing	4672	Special Logon
Audit Success	4/6/2026 1:50:15 AM	Microsoft Windows security auditing	4624	Logon
Audit Success	4/6/2026 1:31:11 AM	Microsoft Windows security auditing	4672	Special Logon
Audit Success	4/6/2026 1:31:11 AM	Microsoft Windows security auditing	4624	Logon
Audit Success	4/6/2026 1:27:49 AM	Microsoft Windows security auditing	4672	Special Logon
Audit Success	4/6/2026 1:27:49 AM	Microsoft Windows security auditing	4624	Logon
Audit Success	4/6/2026 1:27:35 AM	Microsoft Windows security auditing	4672	Special Logon
Audit Success	4/6/2026 1:27:35 AM	Microsoft Windows security auditing	4624	Logon
Audit Success	4/6/2026 1:25:12 AM	Microsoft Windows security auditing	5379	User Account Management
Audit Success	4/6/2026 1:25:11 AM	Microsoft Windows security auditing	5379	User Account Management
Audit Success	4/6/2026 1:25:11 AM	Microsoft Windows security auditing	5379	User Account Management
Audit Success	4/6/2026 1:25:11 AM	Microsoft Windows security auditing	5379	User Account Management
Audit Success	4/6/2026 1:25:11 AM	Microsoft Windows security auditing	5379	User Account Management
Audit Success	4/6/2026 1:25:11 AM	Microsoft Windows security auditing	5379	User Account Management
Audit Success	4/6/2026 1:25:11 AM	Microsoft Windows security auditing	5379	User Account Management
Audit Success	4/6/2026 1:25:11 AM	Microsoft Windows security auditing	5379	User Account Management
Audit Success	4/6/2026 1:25:11 AM	Microsoft Windows security auditing	5379	User Account Management

Event 4624, Microsoft Windows security auditing

General Details

An account was successfully logged on.

Subject:

- Security ID: SYSTEM
- Account Name: DESKTOP-S9IBAA45
- Account Domain: WORKGROUP
- Logon ID: 0x3f7

Logon Information:

- Logon Type: 5
- Process Name: Restricted Admin Module

Log Name: Security

Source: Microsoft Windows security

Event ID: 4624

Level: Information

User: N/A

OpCode: Info

More Information: [Event Log Online Help](#)

Task Category: Logon

Keywords: Audit Success

Computer: DESKTOP-S9IBAA45

[illegible]

7. Analysis

The login activity shows a successful login for the Administrator account using Logon Type 5 (Remote Desktop) at 01:30 PM from IP 10.0.2.15.

This behavior is unusual as administrative access occurred outside normal working hours and originated from an unfamiliar IP address, indicating potential unauthorized access.

8. MITRE ATT&CK Mapping

T1078 → Valid Accounts

T1021 → Remote Services (if RDP used)

Conclusion:

The analysis identified suspicious successful login activity that may indicate compromised credentials and unauthorized remote access. Further investigation and account validation are recommended.

Recommendations:

- Enable multi-factor authentication (MFA)
 - Restrict RDP access
 - Monitor privileged accounts
 - Implement alerting for abnormal login times
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Findings

Repeated failed login attempts detected
Same user targeted multiple times
Pattern indicates suspicious activity

Recommendations

- Implement account lockout policies
 - Use strong passwords
 - Enable multi-factor authentication
 - Monitor login activity regularly
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Conclusion

The analysis suggests a potential brute-force attack based on repeated failed login attempts. Continuous monitoring and preventive controls are recommended
